# Multivariate Data Analysis Problem Bank: 46 Problems

Proof-oriented exam preparation (Chs. 1–11)

Problems marked "2024 Midterm" are from the 2024 Fall midterm (closed-book; points in brackets) and are placed with the chapter they test.
Detailed solutions are provided in a separate file.

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## Part I. Distributions and Inference

## Problem 1 [Ch. 1 — centering matrix and the rank of $\mathbf{S}$]

$\mathbf{H}_n = \mathbf{I}_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top$, $\mathbf{X} \in \mathbb{R}^{n\times p}$.

(a) Show that $\mathbf{H}_n$ is symmetric and idempotent, $\mathbf{H}_n\mathbf{1}_n = \mathbf{0}$ and rank$\mathbf{H}_n = n - 1$.

(b) Show $(n-1)\mathbf{S} = \mathbf{X}^\top\mathbf{H}_n\mathbf{X}$ and prove $\operatorname{rank}(\mathbf{S}) \le \min(n-1, p)$.

(c) Show that $\mathbf{S}$ is singular when $p \ge n$ and explain the consequences for sphering and Mahalanobis distance.

## Problem 2 [Ch. 1 — geometry of a small data set]

$\mathbf{x}_1 = (3, 7)^\top$, $\mathbf{x}_2 = (5, 9)^\top$, $\mathbf{x}_3 = (4, 4)^\top$, $\mathbf{x}_4 = (8, 12)^\top$.

(a) Find $\bar{\mathbf{x}}$, $\mathbf{S}$, $\mathbf{R}$.

(b) Find the eigenvalues and eigenvectors of $\mathbf{S}$ and describe the axes and half-lengths of the data ellipse $(\mathbf{x} - \bar{\mathbf{x}})^\top \mathbf{S}^{-1} (\mathbf{x} - \bar{\mathbf{x}}) = c$.

(c) Compute $d_i^2 = (\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{S}^{-1} (\mathbf{x}_i - \bar{\mathbf{x}})$. Prove that $\sum_{i=1}^{n} d_i^2 = (n-1)p$ in general and check it.

(d) Find the sphering matrix $\mathbf{S}^{-1/2}$.

## Problem 3 [Ch. 1 — rank deficiency]

Three variables $x_1, x_2, x_3 = x_1 + x_2$ are measured on $n > 3$ items.

(a) Find $\mathbf{a} \neq \mathbf{0}$ with $\mathbf{a}^\top(\mathbf{x}_i - \bar{\mathbf{x}}) = 0$ for all $i$, and show $\mathbf{S}\mathbf{a} = \mathbf{0}$.

(b) Relate the deviation vectors $\mathbf{d}_1, \mathbf{d}_2, \mathbf{d}_3$ and show $\operatorname{rank}(\mathbf{S}) \leq 2$.

(c) What is the variance of the third sample principal component?

## Problem 4 [2024 Midterm Q1 (30 pts) — data matrix and a singular covariance]

A bivariate population has mean $\boldsymbol{\mu} = (1, 0)^\top$ and covariance

$$\boldsymbol{\Sigma} = \begin{pmatrix} 4 & -6 \\ -6 & 9 \end{pmatrix}.$$

Let $\mathbf{x}_1, \ldots, \mathbf{x}_n$ be a random sample of size $n = 100$ and $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_n]^\top$.

(a) How many rows and columns does $\mathbf{X}$ have?

(b) Express $\bar{\mathbf{x}}$ and $\mathbf{S}$ using $\mathbf{X}$ only.

(c) Find rank($\mathbf{S}$) (with proof) and describe the spread of these bivariate data.

(d) What is the Mahalanobis distance of $\mathbf{y} = (1, 1)^\top$ from the "center" of this distribution?

(e) If the population is multivariate normal, find the distribution of $\bar{\mathbf{x}}$.

## Problem 5 [Ch. 1, 3 — singular population covariance (proof)]

Let $\mathbf{X}$ have mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$ with $\boldsymbol{\Sigma}\mathbf{a} = \mathbf{0}$ for some $\mathbf{a} \neq \mathbf{0}$.

(a) Prove that $\mathbf{a}^\top \mathbf{X} = \mathbf{a}^\top \boldsymbol{\mu}$ with probability one, i.e. $\mathbf{X}$ lies on a hyperplane.

(b) For a sample $\mathbf{x}_1, \ldots, \mathbf{x}_n$ from this population prove $\mathbf{S}\mathbf{a} = \mathbf{0}$ a.s. and hence $\operatorname{rank}(\mathbf{S}) \leq \operatorname{rank}(\boldsymbol{\Sigma})$ a.s.

(c) Explain why the Mahalanobis distance $(\mathbf{y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{y} - \boldsymbol{\mu})$ is undefined, and what can be said about points $\mathbf{y}$ with $\mathbf{a}^\top (\mathbf{y} - \boldsymbol{\mu}) \neq 0$.

## Problem 6 [2024 Midterm Q2 (10 pts) — extremes of a quadratic form]

Find the maximum and minimum values of $4x_1^2 + 4x_2^2 + 6x_1x_2$ over all $\mathbf{x}^\top = [x_1, x_2]$ with $\mathbf{x}^\top\mathbf{x} = 1$.

## Problem 7 [Ch. 2 — an elementary inequality and a third MLE proof]

(a) For $\mathbf{A} \succ 0$ $(p \times p)$ show $\log|\mathbf{A}| \le \operatorname{tr}(\mathbf{A}) - p$ with equality iff $\mathbf{A} = \mathbf{I}_p$.

(b) For $\mathbf{S}_0 \succ 0$ and $f(\boldsymbol{\Sigma}) = \log|\boldsymbol{\Sigma}| + \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{S}_0)$, show $f(\boldsymbol{\Sigma}) - f(\mathbf{S}_0) = \operatorname{tr}(\mathbf{B}) - \log|\mathbf{B}| - p$ with $\mathbf{B} = \mathbf{S}_0^{1/2}\boldsymbol{\Sigma}^{-1}\mathbf{S}_0^{1/2}$, and deduce $\hat{\boldsymbol{\Sigma}} = \mathbf{S}_0$.

## Problem 8 [Ch. 2 — block-matrix formulas]

(a) Show $|\mathbf{I}_p + \mathbf{u}\mathbf{v}^\top| = 1 + \mathbf{v}^\top\mathbf{u}$ and deduce $|\mathbf{A} + \mathbf{u}\mathbf{v}^\top| = |\mathbf{A}|(1 + \mathbf{v}^\top\mathbf{A}^{-1}\mathbf{u})$.

(b) For symmetric $\mathbf{A} \succ 0$ partitioned as $\begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}$, show $(\mathbf{A}^{-1})_{11} = (\mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21})^{-1}$ by block elimination.

(c) For $\mathbf{W} \succ 0$ show $\dfrac{|\mathbf{W}|}{|\mathbf{W} + \mathbf{b}\mathbf{b}^\top|} = \dfrac{1}{1 + \mathbf{b}^\top\mathbf{W}^{-1}\mathbf{b}}$.

## Problem 9 [Ch. 3 — linear transformations and conditional distributions]

$\mathbf{X} \sim N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\boldsymbol{\mu} = (1, 2, 0)^\top$, $\boldsymbol{\Sigma} = \begin{pmatrix} 4 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{pmatrix}$.

(a) Find the distribution of $X_1 \mid (X_2, X_3) = (x_2, x_3)$.

(b) Find the distribution of $(X_1 + X_2,\ X_1 - X_3)^\top$.

(c) Are $X_1$ and $X_3$ independent? $(X_1, X_2)$ and $X_3$? $X_1$ and $X_2 - \frac{1}{4}X_1$?

## Problem 10 [Ch. 3 — BLP, multiple and partial correlation]

$(Y, X_1, X_2)$ has mean $(5, 1, 2)$ and covariance $\begin{pmatrix} 10 & 3 & 4 \\ 3 & 2 & 1 \\ 4 & 1 & 4 \end{pmatrix}$.

(a) Find $\mathrm{BLP}(Y \mid X_1, X_2)$ and the residual variance.

(b) Find $\rho_{\mathrm{mult}} = \mathrm{mcorr}(\mathbf{X}, Y)$.

(c) Find the partial covariance and partial correlation of $Y$ and $X_1$ adjusted for $X_2$, and compare with the marginal correlation.

(d) For scalar $Z$, prove $\rho_{YX \cdot Z} = \dfrac{\rho_{YX} - \rho_{YZ}\rho_{XZ}}{\sqrt{(1-\rho_{YZ}^2)(1-\rho_{XZ}^2)}}$ and check (c).

## Problem 11 [2024 Midterm Q4 (20 pts) — a recursive (triangular) model]

$X_1 = \epsilon_1$, $X_2 = \beta_{12}X_1 + \epsilon_2$, $X_3 = \beta_{13}X_1 + \beta_{23}X_2 + \epsilon_3$, with $\epsilon_1, \epsilon_2, \epsilon_3 \sim N(0, \sigma^2)$ independent.

(a) Find the conditional variance $\mathrm{Var}(X_1 \mid X_2)$.

(b) Express the inverse covariance matrix of $\mathbf{X} = (X_1, X_2, X_3)$ in terms of $\beta_{12}, \beta_{13}, \beta_{23}, \sigma^2$.

## Problem 12 [Ch. 3 — the precision matrix (proof)]

Let $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\boldsymbol{\Omega} = \boldsymbol{\Sigma}^{-1} = (\omega_{ij})$. Write $\mathbf{X}_{-i}$ for all components except $X_i$.

(a) Prove $\mathrm{Var}(X_i \mid \mathbf{X}_{-i}) = 1/\omega_{ii}$ and $\mathrm{E}(X_i \mid \mathbf{X}_{-i}) = \mu_i - \frac{1}{\omega_{ii}} \sum_{j \neq i} \omega_{ij}(x_j - \mu_j)$.

(b) Prove $\mathrm{pcorr}(X_i, X_j; \text{rest}) = -\dfrac{\omega_{ij}}{\sqrt{\omega_{ii}\omega_{jj}}}$, so $X_i \perp\!\!\!\perp X_j \mid \text{rest} \iff \omega_{ij} = 0$.

(c) Use (a)–(b) to interpret the zero pattern of the precision matrix in Problem 11.

## Problem 13 [Ch. 3 — uncorrelated normal marginals that are not independent]

$X_1 \sim N(0,1)$, $W \perp\!\!\!\perp X_1$ with $P(W = \pm 1) = \frac{1}{2}$, $X_2 = WX_1$.

(a) Show $X_2 \sim N(0,1)$ and $\text{Cov}(X_1, X_2) = 0$.

(b) Show $X_1$ and $X_2$ are not independent.

(c) Show $(X_1, X_2)$ is not bivariate normal using a linear combination. Why is there no contradiction with Theorem 3.9(b)?

## Problem 14 [Ch. 3 — quadratic forms]

(a) If $\mathbf{Y} \sim N_n(\mu \mathbf{1}_n, \sigma^2 \mathbf{I}_n)$, show $(n-1)S^2/\sigma^2 = \mathbf{Y}^\top \mathbf{H}_n \mathbf{Y}/\sigma^2 \sim \chi^2_{n-1}$ and $\bar{Y} \perp\!\!\!\perp S^2$ using Theorem 3.20 and Theorem 3.23.

(b) $\mathbf{X} \sim N_2(\mathbf{0}, \mathbf{\Sigma})$, $\mathbf{\Sigma} = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$, $|\rho| < 1$. For which $\rho$ is $X_1^2 + X_2^2$ chi-square? What is the law of $\frac{1}{1-\rho^2}(X_1^2 - 2\rho X_1 X_2 + X_2^2)$?

## Problem 15 [2024 Midterm Q5 (30 pts) — prove in detail]

(a) Let $\mathbf{X}_1, \ldots, \mathbf{X}_n$ be a random sample from a distribution with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$. Then $\mathbf{S}_n = \frac{1}{n}\sum_{i=1}^{n}(\mathbf{X}_i - \bar{\mathbf{X}})(\mathbf{X}_i - \bar{\mathbf{X}})^\top$ is biased with $\mathrm{E}(\mathbf{S}_n) = \frac{n-1}{n}\boldsymbol{\Sigma}$.

(b) Let $\mathbf{X} \sim N_p(\mathbf{0}, \sigma^2\mathbf{I})$ and $\mathbf{A}$ symmetric $p \times p$. Then $Y = \mathbf{X}^\top\mathbf{A}\mathbf{X}/\sigma^2 \sim \chi^2(m)$ iff $\mathbf{A}$ is idempotent of rank $m \le p$. [Hint: you may use without proof: $\mathbf{A} = \boldsymbol{\Gamma}\boldsymbol{\Lambda}\boldsymbol{\Gamma}^\top$ for symmetric $\mathbf{A}$; a symmetric idempotent matrix has eigenvalues 0 or 1 and $\mathrm{rank}(\mathbf{A}) = \#\{\text{nonzero eigenvalues}\} = \mathrm{tr}(\mathbf{A})$.]

(c) Let $\mathbf{X} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$, $\boldsymbol{\Sigma} \succ 0$, $\mathbf{A}$ symmetric. Then $Y = \mathbf{X}^\top\mathbf{A}\mathbf{X} \sim \chi^2(m)$ iff either (i) $\mathbf{A}\boldsymbol{\Sigma}$ or (ii) $\boldsymbol{\Sigma}\mathbf{A}$ is idempotent of rank $m$. [Hint: you may use (b).]

## Problem 16 [Ch. 4 — Wishart moments]

$\mathbf{M} \sim W_p(m, \mathbf{\Sigma})$.

(a) For constant $\mathbf{a} \neq \mathbf{0}$ show $\mathbf{a}^\top \mathbf{M}\mathbf{a} / \mathbf{a}^\top \mathbf{\Sigma}\mathbf{a} \sim \chi^2_m$ and $\mathrm{Var}(\mathbf{a}^\top \mathbf{M}\mathbf{a}) = 2m(\mathbf{a}^\top \mathbf{\Sigma}\mathbf{a})^2$.

(b) Check consistency with $\mathrm{Var}(M_{ij}) = m(\Sigma_{ij}^2 + \Sigma_{ii}\Sigma_{jj})$ for $i = j$.

## Problem 17 [Ch. 4 — the inverse-Wishart ratio theorem (proof)]

Let $\mathbf{M} \sim W_p(m, \mathbf{\Sigma})$, $\mathbf{\Sigma} \succ 0$, $m > p-1$. Prove in detail that $(\mathbf{\Sigma}^{-1})_{11}/(\mathbf{M}^{-1})_{11} \sim \chi^2_{m-p+1}$ and that it is independent of the last $p-1$ columns of the underlying data matrix. Identify where the degrees of freedom $m-p+1$ come from.

## Problem 18 [Ch. 4 — inverse-Wishart ratios and $\mathrm{E}(\mathbf{S}^{-1})$]

$\mathbf{X}_1, \ldots, \mathbf{X}_n \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $n > p + 2$.

(a) For constant $\mathbf{a} \neq \mathbf{0}$ show $(n-1)\dfrac{\mathbf{a}^\top \boldsymbol{\Sigma}^{-1}\mathbf{a}}{\mathbf{a}^\top \mathbf{S}^{-1}\mathbf{a}} \sim \chi^2_{n-p}$.

(b) Using $\mathrm{E}(1/V) = 1/(k-2)$ for $V \sim \chi^2_k$, show $\mathrm{E}(\mathbf{S}^{-1}) = \frac{n-1}{n-p-2}\boldsymbol{\Sigma}^{-1}$.

## Problem 19 [Ch. 4 — constrained MLEs]

For $\mathbf{X}_1, \ldots, \mathbf{X}_n \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ find the MLE of

(a) $\boldsymbol{\Sigma}$ when $\boldsymbol{\mu} = \boldsymbol{\mu}_0$ is known;

(b) $\sigma^2$ when $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}_p$;

(c) $\boldsymbol{\Sigma}$ when it is diagonal.

## Problem 20 [Ch. 4 — large-sample theory]

$\mathbf{X}_i$ iid with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma} \succ 0$ (not normal).

(a) Show $n(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top \mathbf{S}^{-1}(\bar{\mathbf{X}} - \boldsymbol{\mu}) \Rightarrow \chi^2_p$ via CLT, Slutsky and CMT.

(b) For $p = 2$, $\mu_2 \neq 0$, find the limit law of $\sqrt{n}(\bar{X}_1/\bar{X}_2 - \mu_1/\mu_2)$ by the delta method.

## Problem 21 [Ch. 4 — asymptotics of sample covariances (proof)]

Let $\mathbf{X}_i \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

(a) Prove $\sqrt{n}(s_{jk} - \sigma_{jk}) \Rightarrow N(0, \sigma_{jk}^2 + \sigma_{jj}\sigma_{kk})$.

(b) Deduce an approximate 95% confidence interval for $\sigma_{jj}$ and, by the delta method, for $\log \sigma_{jj}$.

## Problem 22 [Ch. 5 — computing a $T^2$ test]

$n = 4$, $\mathbf{X} = \begin{pmatrix} 2 & 12 \\ 8 & 9 \\ 6 & 9 \\ 8 & 10 \end{pmatrix}$, $H_0 : \boldsymbol{\mu} = (7, 11)^\top$, $\alpha = 0.05$.

(a) Find $\bar{\mathbf{x}}$, $\mathbf{S}$, $\mathbf{S}^{-1}$ and $T^2$; test $H_0$ and give the $p$-value.

(b) Compute Wilks' $\Lambda^{2/n} = |\hat{\boldsymbol{\Sigma}}|/|\hat{\boldsymbol{\Sigma}}_0|$ directly and check $(1 + T^2/(n-1))^{-1}$.

(c) Give 95% $T^2$ simultaneous and Bonferroni intervals for $\mu_1, \mu_2$ and compare.

## Problem 23 [Ch. 5 — likelihood ratio tests for the mean]

(a) Prove $\Lambda^{2/n} = (1 + T^2/(n-1))^{-1}$ by the determinant lemma.

(b) Find $-2\log\Lambda$ for $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$ when $\boldsymbol{\Sigma}$ is known and give its exact distribution.

(c) When $\boldsymbol{\Sigma}$ is unknown, show $-2\log\Lambda \Rightarrow \chi^2_p$ from (a).

## Problem 24 [Ch. 5 — likelihood ratio test for $H_0 : \boldsymbol{\Sigma} = \boldsymbol{\Sigma}_0$ (proof)]

$\mathbf{X}_1, \ldots, \mathbf{X}_n \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\boldsymbol{\mu}$ unknown.

(a) Derive $W = -2 \log \Lambda = -n \log |\boldsymbol{\Sigma}_0^{-1} \mathbf{S}_0| - np + n\text{tr}(\boldsymbol{\Sigma}_0^{-1} \mathbf{S}_0)$.

(b) Show $W = n \sum_i (l_i - \log l_i - 1) \geq 0$, $l_i$ the eigenvalues of $\boldsymbol{\Sigma}_0^{-1} \mathbf{S}_0$, with equality iff $\mathbf{S}_0 = \boldsymbol{\Sigma}_0$.

(c) For $p = 1$ prove directly that $W \Rightarrow \chi^2_1$ under $H_0$.

(d) Explain why the degrees of freedom are $p(p+1)/2$ in general.

## Problem 25 [Ch. 5 — simultaneous intervals are shadows of the ellipsoid]

$E = \{\boldsymbol{\mu} : n(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top \mathbf{S}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}) \le c^2\}$.

(a) Show $\max_{\boldsymbol{\mu} \in E} \mathbf{a}^\top \boldsymbol{\mu} = \mathbf{a}^\top \bar{\mathbf{x}} + c\sqrt{\mathbf{a}^\top \mathbf{S}\mathbf{a}/n}$ (and symmetrically for the minimum).

(b) Deduce that the event "all $T^2$ intervals of Theorem 5.12 hold" is exactly $\{\boldsymbol{\mu} \in E\}$.

## Problem 26 [Ch. 5 — Bonferroni versus $T^2$]

For $n = 20$, $p = 3$, $\alpha = 0.05$ find the critical values of (i) one-at-a-time $t$, (ii) Bonferroni and (iii) $T^2$ intervals for the three component means, and the length ratio Bonferroni/$T^2$. When is each used?

## Problem 27 [Ch. 6 — repeated measures design]

Six subjects received all three treatments:

| | | | | | | |
|---|---|---|---|---|---|---|
| Treatment 1 | 10 | 8 | 11 | 9 | 12 | 10 |
| Treatment 2 | 12 | 11 | 13 | 9 | 15 | 13 |
| Treatment 3 | 15 | 11 | 16 | 12 | 17 | 14 |

Test $H_0 : \mu_1 = \mu_2 = \mu_3$ at $\alpha = 0.05$ with $\mathbf{C} = \begin{pmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$, and check it equals a one-sample $T^2$ on $\mathbf{D}_j = \mathbf{C}\mathbf{y}_j$.

## Problem 28 [Ch. 6 — two-sample $T^2$]

$n_1 = n_2 = 50$, $\bar{\mathbf{x}}_1 = (8.3, 4.1)^\top$, $\bar{\mathbf{x}}_2 = (10.2, 3.9)^\top$, $\mathbf{S}_{\text{pooled}} = \begin{pmatrix} 2 & 1 \\ 1 & 5 \end{pmatrix}$.

(a) Test $H_0 : \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2$ at $\alpha = 0.05$.

(b) Give 95% simultaneous componentwise intervals; which component drives the difference?

(c) Find Wilks' $\Lambda^*$ of the one-way MANOVA ($g = 2$) from $T^2$.

## Problem 29 [Ch. 6 — one-way MANOVA by hand]

| | |
|---|---|
| Group 1 | $(9,3),(6,2),(9,7)$ |
| Group 2 | $(0,4),(2,0)$ |
| Group 3 | $(3,8),(1,9),(2,7)$ |

Find $\mathbf{B}$, $\mathbf{W}$, $\Lambda^*$; test at $\alpha = 0.01$ with the exact $F$ ($p = 2$) and with Bartlett's approximation; recompute $\Lambda^*$ from the eigenvalues of $\mathbf{W}^{-1}\mathbf{B}$.

## Problem 30 [Ch. 6 — expectations of $\mathbf{B}$ and $\mathbf{W}$]

In the one-way MANOVA model show $\mathrm{E}\mathbf{W} = (n-g)\boldsymbol{\Sigma}$ and $\mathrm{E}\mathbf{B} = (g-1)\boldsymbol{\Sigma} + \sum_{\ell} n_\ell \boldsymbol{\tau}_\ell \boldsymbol{\tau}_\ell^{\top}$, and explain how this supports Wilks' test.

## Problem 31 [Ch. 6 — Wilks' lambda is the LRT (proof)]

(a) Derive the MLEs under $H_0 : \boldsymbol{\tau}_1 = \cdots = \boldsymbol{\tau}_g = \mathbf{0}$ and under $H_1$, and show $\Lambda^{2/n} = |\mathbf{W}|/|\mathbf{B}+\mathbf{W}|$.

(b) Explain the degrees of freedom $p(g-1)$ in Bartlett's approximation.

(c) For $g = 2$ prove $\Lambda^* = (1 + T^2/(n-2))^{-1}$ and derive the third row of the Wilks table.

## Problem 32 [Ch. 6 — two populations with unequal covariances]

$\mathbf{X}_{1j} \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)$, $\mathbf{X}_{2j} \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2)$ independent, $n_1, n_2 \to \infty$, $n_1/(n_1+n_2) \to \kappa \in (0,1)$. Show $[\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)]^\top (\frac{1}{n_1}\mathbf{S}_1 + \frac{1}{n_2}\mathbf{S}_2)^{-1}[\cdots] \Rightarrow \chi^2_p$.

### Part II. Dimension Reduction, Classification, Clustering

## Problem 33 [Ch. 7 — computing principal components]

$$\boldsymbol{\Sigma} = \begin{pmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

(a) Find the PCs, their variances and proportions of variance.

(b) Find $\mathrm{Corr}(Z_1, X_1)$ and $\mathrm{Corr}(Z_1, X_2)$.

(c) How does correlation PCA differ?

## Problem 34 [2024 Midterm Q3 (10 pts) — PCA of a singular covariance]

$Y_1, Y_2$ have covariance $\mathbf{\Sigma} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$. Calculate the proportion of the total population variance explained by the first principal component.

## Problem 35 [Ch. 7 — equicorrelation]

$\boldsymbol{\Sigma} = \sigma^2\{(1-\rho)\mathbf{I}_p + \rho\mathbf{1}\mathbf{1}^\top\}$, $0 < \rho < 1$. Find the eigenvalues and eigenvectors, interpret PC1 and its proportion of variance, and discuss $\rho \to 1$.

## Problem 36 [Ch. 7 — reconstruction error and SVD]

(a) For orthonormal $\mathbf{V} \in \mathbb{R}^{p\times q}$ show $\sum_i \|(\mathbf{x}_i - \bar{\mathbf{x}}) - \mathbf{V}\mathbf{V}^\top(\mathbf{x}_i - \bar{\mathbf{x}})\|^2 = (n-1)\{\mathrm{tr}\mathbf{S} - \mathrm{tr}(\mathbf{V}^\top\mathbf{S}\mathbf{V})\}$, and with von Neumann's inequality show the minimum is $(n-1)\sum_{j>q}\hat{\lambda}_j$.

(b) If $\tilde{\mathbf{X}}^\top = \mathbf{U}\mathbf{D}\mathbf{V}^\top$ (SVD), show the PC score matrix is $\mathbf{D}\mathbf{V}^\top$.

## Problem 37 [Ch. 8 — computing canonical correlations]

$\mathbf{R}_{11} = \begin{pmatrix} 1 & .4 \\ .4 & 1 \end{pmatrix}$, $\mathbf{R}_{22} = \begin{pmatrix} 1 & .2 \\ .2 & 1 \end{pmatrix}$, $\mathbf{R}_{12} = \begin{pmatrix} .5 & .6 \\ .3 & .4 \end{pmatrix}$. Find $\rho_1^*, \rho_2^*$ and the first canonical coefficients, and check $\prod(1 - \rho_k^{*2}) = |\mathbf{R}|/(|\mathbf{R}_{11}||\mathbf{R}_{22}|)$.

## Problem 38 [Ch. 8 — properties of canonical correlations]

(a) For $q = 1$ show $\rho_1^2$ is the squared multiple correlation.

(b) Show invariance under non-singular $\mathbf{X} \mapsto \mathbf{AX} + \mathbf{b}$, $\mathbf{Y} \mapsto \mathbf{CY} + \mathbf{d}$.

(c) Show $\mathrm{Corr}(\xi_k, \omega_l) = 0$ for $k \neq l$ using the SVD.

## Problem 39 [Ch. 9 — the factor model]

(a) For $\boldsymbol{\Lambda} = \begin{pmatrix} 4 & 1 \\ 7 & 2 \\ -1 & 6 \\ 1 & 8 \end{pmatrix}$, $\boldsymbol{\Psi} = \mathrm{diag}(2, 4, 1, 3)$ find $\boldsymbol{\Sigma}$ and the communalities; show $\boldsymbol{\Lambda}\mathbf{T}$ gives the same $\boldsymbol{\Sigma}$ for orthogonal $\mathbf{T}$.

(b) For $p = 3$, $k = 1$, $\mathbf{R} = \begin{pmatrix} 1 & .9 & .7 \\ .9 & 1 & .4 \\ .7 & .4 & 1 \end{pmatrix}$, find $\lambda_i, \psi_i$ reproducing $\mathbf{R}$ exactly. What goes wrong (Heywood case)?

(c) Derive $\nu = \{(p-k)^2 - p - k\}/2$.

## Problem 40 [Ch. 9 — factor scores]

Derive the WLS score $\hat{\mathbf{F}}^{\text{wls}} = (\boldsymbol{\Lambda}^\top \boldsymbol{\Psi}^{-1} \boldsymbol{\Lambda})^{-1} \boldsymbol{\Lambda}^\top \boldsymbol{\Psi}^{-1} (\mathbf{x} - \boldsymbol{\mu})$, prove $\hat{\mathbf{F}}^{\text{reg}} = (\mathbf{I} + \boldsymbol{\Delta}^{-1})^{-1} \hat{\mathbf{F}}^{\text{wls}}$, and show $\mathrm{E}[\hat{\mathbf{F}}^{\text{wls}} \mid \mathbf{F}] = \mathbf{F}$.

## Problem 41 [Ch. 10 — LDA by hand]

$\boldsymbol{\mu}_1 = (1, 2)^\top$, $\boldsymbol{\mu}_2 = (3, 1)^\top$, $\boldsymbol{\Sigma} = \begin{pmatrix} 2 & .5 \\ .5 & 1 \end{pmatrix}$, $\pi_1 = .6$, $\pi_2 = .4$.

(a) Classify $\mathbf{x}_0 = (2, 2)^\top$ by LDA and find $P(Y = 1 \mid \mathbf{x}_0)$.

(b) With equal priors prove the error rate is $\Phi(-\Delta/2)$, $\Delta^2 = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top \boldsymbol{\Sigma}^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$, and compute it.

(c) Find the error rate with the given priors.

## Problem 42 [Ch. 10 — Fisher's discriminant and QDA]

(a) Show the maximizer of $\frac{(\mathbf{u}^\top \mathbf{d})^2}{\mathbf{u}^\top \mathbf{S}_p \mathbf{u}}$ is $\mathbf{u} \propto \mathbf{S}_p^{-1}\mathbf{d}$ and find the maximum.

(b) For $f_1 = N(0,1)$, $f_2 = N(0,4)$, $\pi_1 = \pi_2$ find the Bayes rule. Why is the boundary not linear?

## Problem 43 [Ch. 10 — generalized Rayleigh quotient (proof)]

Let $\mathbf{A} \succeq 0$, $\mathbf{B} \succ 0$ be $p \times p$.

(a) Prove $\max_{\mathbf{u} \neq \mathbf{0}} \frac{\mathbf{u}^\top \mathbf{A}\mathbf{u}}{\mathbf{u}^\top \mathbf{B}\mathbf{u}} = \lambda_1(\mathbf{B}^{-1}\mathbf{A})$ and identify the maximizer.

(b) Show the successive maximizers subject to $\mathbf{u}^\top \mathbf{B}\mathbf{u}_j = 0$ are the further eigenvectors of $\mathbf{B}^{-1}\mathbf{A}$.

(c) For Fisher's multi-group discriminant ($\mathbf{A} = \mathbf{B}_{\text{MANOVA}}$, $\mathbf{B} = \mathbf{W}$) show there are at most $\min(p, K-1)$ nonzero eigenvalues.

## Problem 44 [Ch. 11 — $K$-means and one EM step]

(a) Cluster $A(5,3)$, $B(-1,1)$, $C(1,-2)$, $D(-3,-2)$ by $K$-means from the initial clusters $\{A,B\},\{C,D\}$, giving $W(C)$ at each step.

(b) Fit $\pi_1\phi(x;\mu_1,1)+(1-\pi_1)\phi(x;\mu_0,1)$ (unit variance fixed) to $\{-2,-1,0,4,5,6\}$. From $\pi_1=.5$, $\mu_0=0$, $\mu_1=1$ do one EM step and check that the log-likelihood increases.

## Problem 45 [Ch. 11 — EM monotonicity and $K$-means as a limit (proof)]

(a) Prove $\ell(\boldsymbol{\theta}^{(m+1)}) \geq \ell(\boldsymbol{\theta}^{(m)})$ for the EM algorithm.

(b) For the mixture $\sum_k \pi_k \phi(\mathbf{x}; \boldsymbol{\mu}_k, \sigma^2 \mathbf{I})$ with equal $\pi_k$, show that as $\sigma \to 0$ the E-step responsibilities become hard assignments to the nearest mean and the M-step becomes the $K$-means centroid update.

## Problem 46 [Ch. 11 — hierarchical clustering]

For the distance matrix

$$\mathbf{D} = \begin{pmatrix} 0 & & & & \\ 9 & 0 & & & \\ 3 & 7 & 0 & & \\ 6 & 5 & 9 & 0 & \\ 11 & 10 & 2 & 8 & 0 \end{pmatrix}$$

apply single, complete and average linkage and give the merge order and heights.